

COMPUTATIONAL MATERIALS PHYSICS & DFT

Research Progress and Electronic Structure Modeling

Presenter: Research Group

Institution: Universidade Federal do ABC (UFABC)

Project: Advanced Ab Initio Simulations

Date: 2026/09/30



Introduction & Methodology

Summary & Next Steps

Density Functional Theory and Modern Electronic Calculations

Kohn-Sham Equations

Self-consistent single-particle formulation:

$$\left[-\frac{1}{2}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + V_{\text{H}}(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$$

- ⊙ Projected Augmented Wave (PAW) method
- ⊙ Generalized Gradient Approximation (PBE)
- ⊙ On-site Coulomb interaction (+ U)

Simulation Protocol

- ⊙ Geometry relaxation ($F_{\text{max}} < 0.01 \text{ eV/\AA}$)
- ⊙ Electronic convergence (10^{-6} eV)
- ⊙ Micro-batch Slurm execution on HPC clusters

UFABC HPC High-Performance Cluster

Summary and Next Steps

Conclusions and Outlook

Key Milestones

- ⦿ Successfully integrated custom Nord Beamer theme with institutional branding.
- ⦿ High-contrast UFABC grayscale vector-aligned insignia on title slide.
- ⦿ Ready for conference presentations, seminars, and thesis defenses.

Thank you for your attention!

Questions and Discussion